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Application of nanodiamonds in Cu(II)-based rhodamine B probes for NO detection and cell imaging†

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Nitric oxide functions as an important signalling molecule in many biological systems. Nanodiamonds (NDs) have attracted enormous attention in the field of biomaterial science for biosensing applications and drug/gene delivery. In this work, one novel Cu(II)-based rhodamine B probe for NO detection was composed which was covalently bound to the nanodiamond (ND) surface. The results showed that the covalently conjugated ND probe could detect Cu^{2+} and NO sequentially with high sensitivity, high stability and good reusability compared to a free rhodamine B probe in CH_3CN/HEPES (pH 6.8, v/v, 1 : 1). The detection limit for NO was estimated to be 10.5 nM. In addition, the response to NO was reversible in the presence of O_3, which has attracted attention in the fields of biology and environmental science. Further analysis of confocal fluorescence microscopy images and the MTT assay indicated that the conjugated ND probe has favorable biocompatibility and low toxicity. This new Cu^{2+} and NO selective fluorescent ND probe may have potential applications in environmental science and biology.

1. Introduction

Nitric oxide (NO) has been heavily studied over the past two decades due to its multitude of physiological functions and potential therapeutic promise.[1] NO is a free radical gas that functions as an important signaling molecule in many biological systems. It is involved in the regulation of diverse physiological and pathophysiological mechanisms in cardiovascular relaxation, nervous transmission and immunological responses.[2–5] The direct measurement of NO *in vivo* has been rare because of experimental difficulties and the short half-life of NO.[6,7] Therefore, in clinical studies, NO synthesis has been monitored by the detection of nitrite and nitrate levels in biological fluids. To achieve the direct detection of nitric oxide, a successful approach involving the exploration of Cu(II)-based probes for NO detection has been developed.[8–14] These studies formed the basis of an effective strategy for developing metal-based probes for the imaging of NO production in living cells, but these various Cu(II)-based systems were not satisfactory as biological NO sensors. Of major interest is the desire to design NO sensors with good biocompatibility.

Nanodiamonds (NDs), owing to their low toxicity, high chemical stability, high affinity for biomolecules and ease of surface functionalization, have attracted enormous attention in the field of biomaterial science.[15–17] In the context of biological and medical applications, several biocompatibility studies of the nanodiamond particles *in vitro* or *in vivo* have shown that nanodiamond particles are among the least toxic of all carbon-based nano materials tested so far[18,19] and are considered relatively biocompatible in a diverse range of cell types. Surface functionalization of NDs provides the opportunity to graft different functional moieties such as bioactive structures, polymerizable monomers and radical initiators. Their surfaces can be modified either by covalent or non-covalent attachment of organic structures.[20] Numerous studies have revealed that there is increasing interest in utilizing surface-modified NDs for biosensor applications and for optical labeling and drug/gene delivery.[21–24]

The aim of this work was to construct a nanodiamond-conjugated rhodamine fluorescent sensor for Cu(II) which could then be developed as an excellent NO selective fluorescent particle. As classic dyes/fluorophores, rhodamine dyes are widely used as small molecule-based turn-on fluorescent probes in routine optical analysis owing to their high absorption coefficient, fluorescence emission invisible region, and high fluorescence quantum yield.[25–28] Spirolactam-type rhodamine derivatives are nonfluorescent and colorless, whereas spirolactam ring-opening gives rise to strong fluorescence emission. Great progress has been achieved in the development of new luminescent chemodosimeters based

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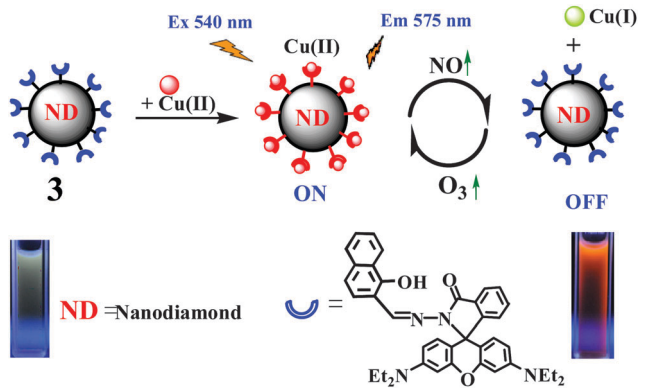


Fig. 1 Proposed recycling mechanism for fluorescence OFF–ON of **3** upon addition of Cu^{2+}, NO and O_3 sequentially.

on this process in the past several years. Cu^{2+} can induce the spirolactam ring-opening process of rhodamine derivatives due to the chelation of metal ions with the probes, which has become the major strategy for designing Cu^{2+}-selective luminescent chemodosimeters with a luminescence turn-on response.[29–31] Even though the stability and cytotoxicity of rhodamine B probes have attracted much attention, low or no cytotoxicity should be taken into account in the design of novel NO probe systems for practical applications. A nanodiamond-conjugated rhodamine fluorescent sensor should be ideal to solve this problem.

In this work, (i) one new Cu(II)-based rhodamine B probe for NO detection was designed which was covalently bound to the ND surface (Fig. 1). The conjugated ND probe **3** had high sensitivity, good biocompatibility and high stability for Cu^{2+} and NO compared to the free rhodamine B derivative. (ii) This probe for NO detection was related to the reduction of Cu(II) to Cu(I). The response to NO was reversible in the presence of a strong oxidant O_3 in CH_3CN–HEPES solution. So the 'OFF–ON' changes of the **3**–Cu^{2+} + NO system may be further developed as the first reversible fluorescent ND probe for O_3.

2. Experimental

2.1 Materials and apparatus

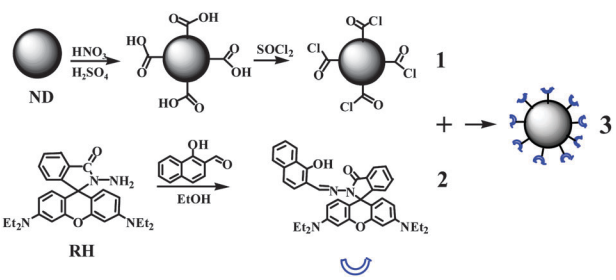
All the reagents were of analytical grade and used as received. Nanodiamonds (non fluorescent) (mean sizes of 140 nm) were donated as a present by Prof. Chang (the Institute of Atomic and Molecular Sciences, Academia Sinica, Taiwan). NO donor DEA·NONOate (CAS: 372965-00-9) was purchased from Sigma. A stock solution of DEA·NONOate was prepared in 0.01 M NaOH solution. The solutions of metal ions Mg^{2+}, Ba^{2+}, Fe^{3+}, Co^{2+}, Ni^{2+}, Mn^{2+}, Cr^{3+}, Al^{3+}, Zn^{2+}, Cu^{2+}, Hg^{2+}, Ag^{+}, Cd^{2+} and Pb^{2+} were prepared from their nitrate salts; the solutions of F^-, Cl^-, Br^-, I^-, CO_3^{2-}, SO_3^{2-}, NO_2^-, S^{2-}, ClO^- anions were prepared from their sodium salts, respectively. Ozone was generated by feeding high purity oxygen into a Welsbach T-series ozone generator. SW480 cells were provided by the Gene Engineering Center of Shanxi University. Dulbecco Minimum Essential Medium (DMEM) was purchased from Thermo Fisher Biological and Chemical Product (Beijing, China). The heat-inactivated fetal bovine serum (FBS) was from GIBCO (Grand Island, NY).

^{1}H NMR spectra were recorded on Bruker-600 MHz spectrometers; the chemical shifts (δ) were reported as ppm in DMF. Fluorescence responses were recorded on a Fluoromax-X spectrofluorometer (HORIBA). UV-visible (UV-Vis) spectra were measured using a Varian 50 BIO spectrophotometer. IR spectra were recorded using a Bruker TENSOR 21 FT-IR spectrophotometer. Elemental analyses were performed on a Vario EL III analyzer. Scanning electron microscopy (SEM) was carried out on a JEOL-JSM-6701F scanning electron microscope at 10 kV. All pH measurements were performed using a pH meter (Mettler Toledo, Switzerland). A water-jacketed CO_2 cell incubator was from Shanghai Lishen Scientific Instruments Co., Ltd. Bioimaging of the sensors was performed by laser confocal fluorescence imaging using a Leica TCS SP5 laser scanning microscope.

2.2 Synthesis of 2 and 3

Rhodamine hydrazide (RH) was synthesised according to the reported method.[32] RH (0.46 g, 1.0 mM) and 1-hydroxy naphthalene formaldehyde (0.18 g, 1.0 mM) were dissolved in 30 mL of ethanol, and the mixture was refluxed for 3 h. After cooling to room temperature, a pink solid was precipitated. The solid was purified by recrystallization to provide **2** as a pink solid (0.32 g, yield 52.5%, Mr = 610). ^{1}H-NMR ($CDCl_3$), δ (ppm, 600 MHz, TMS): δ 9.85 (1H, Phen-H), 8.03 (1H, Phen-H), 7.83 (2H, Phen-H), 7.67 (2H, Phen-H), 7.56 (1H, Phen-H), 7.28 (2H, Phen-H), 7.19 (4H, Phen-H), 7.08 (2H, Phen-H), 3.31 (8H, NCH_2CH$_3$), 1.13 (12H, NCH$_2$$CH_3$) (Fig. S1, ESI†). Anal. calcd for C: 76.70, H: 6.27, N: 7.86; find: C: 76.75, H: 6.54, N: 7.80.

The synthetic route used to obtain the title compound **3** is shown in Scheme 1. Firstly, the carboxylated/oxidized ND (0.2 g) was dried in a vacuum oven at 70 °C overnight and then suspended in 2 mL of anhydrous dimethyl formamide (DMF) in an ultrasonic bath.[33,34] This dispersion was immediately added to 20 mL of thionyl chloride ($SOCl_2$) and refluxed for 24 h at 70 °C. Then **2** (0.1 g) was added into the mixed liquids and refluxed for another 2 h. The mixture was evaporated using a rotary evaporator to remove excess $SOCl_2$ at 40 °C. Third, the nanodiamond suspension was centrifuged at 15 000 rpm for 5 min and washed with 20 mL of CH_2Cl_2 for three cycles to provide **3** (0.15 g). The amount of **2** adsorbed onto the surface of **3** was determined by calculating the change in the concentration of **2** before and after adsorption using a UV-Vis spectrophotometer



Scheme 1 The synthetic route used to obtain compound **3**.

at 555 nm according to the obtained standard equation (1.0 mg **3** is coating with 50 μg **2**). The formation of **3** was characterized by Fourier transform infrared (FTIR) analysis and scanning electron microscopy (SEM) (Fig. S2 and S3, ESI†).

2.3 UV-Vis and fluorescence titration study

The inorganic salt was dissolved in distilled water to afford 1.0 mM solution. The 1.0 mM stock suspension of **3** was prepared in CH_3CN/water (1/1, v/v). The fluorescence spectral properties of sensors (10 μM) were investigated in 0.1 M HEPES buffer at pH 6.8 (CH_3CN/water, 1/1, v/v) as a function of added [Cu^{2+}]. Then various analytes of NO donor DEA·NONOate were added to the solution of the probe **3** + Cu^{2+} system in a 3 mL closed cuvette. The resulting solution was kept at 25 °C for 30 min, and then the fluorescence spectra were recorded. Last, the samples were exposed to ozone (~5 μM) followed by 10 min incubation at room temperature before fluorescence measurement. Fluorescence measurements were carried out with excitation and emission slit widths of 2.5 and 2.5 nm and the PMT voltage (photomultipler tube) and the excitation wavelength were 400 V and 540 nm, respectively. The concentration of **3** (10 μM) refers to the concentration of the attached **2** (10 μM) if not specified.

2.4 Cell culture and fluorescence imaging

SW480 cells were cultured in DMEM (Dulbecco's modified Eagle's medium) supplemented with 10% FBS, and 1% penicillin–streptomycin at 37 °C in a humidified atmosphere with 5% CO_2 and 95% air. The cells were cultured to confluence with medium exchanges every other day. Before the imaging experiment, the cells were seeded in 24-well plates for 24 h. Powders **3** were resuspended in phosphate buffered saline (PBS) solution for cell experiments. The cells were incubated with **3** (10 μM) at 37 °C under 5% CO_2 for 60 min, washed 3 times to remove the remaining probe and bathed in DMEM containing no FBS prior to imaging. Then 15 μM Cu^{2+} was added in the growth medium for 0.5 h at 37 °C, washed 3 times with PBS (phosphate buffered saline) buffer. Cells were imaged under a laser scanning confocal microscope. Then, the cells were incubated with the NO donor DEA·NONOate for 30 min in DMEM. After each treatment, the cells were washed with PBS for 3 times. Emission was collected at 580–630 nm for the red channel (excited at 543 nm).

2.5 Cellular cytotoxicity

Cellular cytotoxicity was carried out using the traditional method.[23] Simply, SW480 cells (5×10^3) were plated into 96 well microtiter plates (Solarbio). After 16 h, the cells were exposed to ND, **2** and **3** (10 μM) in DMEM with 10% FBS for 12 h. Then the solutions were aspirated and replaced by DMEM (180 μL), followed by the addition of 5 mg mL^{-1} MTT (methylthiazolyldiphenyl-tetrazolium bromide) solution (final concentration of 0.5 mg mL^{-1}) and incubated for 4 h. The unreacted dye was removed by aspiration. Each group was a set of six samples. The data were obtained from three independent experiments. The absorbance for MTT analysis was recorded on a multiplate reader (Tecan, safire 2) at 490 nm. The spectrophotometer was calibrated to zero absorbance using culture medium without cells. The relative cell viability (%) is related to the control cell viability = A_{490} (sample)/A_{490} (control) × 100%, where A_{490} (sample) is the absorbance value of NDs, **2** and **3** treated cells respectively, and A_{490} (control) is the absorbance value of cells as control groups.

3. Results and discussion

3.1 Cu^{2+} detection by probe 3

First, the selectivity of the ND probe **3** for Cu^{2+} was observed in the fluorescence spectra. The fluorescence response of **3** to various cations and its selectivity for Cu^{2+} are illustrated in Fig. 2a. It was clear that **3** showed very weak fluorescence in the absence of metal ions in CH_3CN–HEPES buffer (1 : 1, v/v). However, the addition of Cu^{2+} (2.0 equiv.) resulted in a remarkably enhanced fluorescence intensity (λ_{ex} = 540 nm).

It was visualised by a change in the solution colour from colourless to orange under illumination using a 365 nm UV lamp. Meanwhile, no obvious responses could be observed upon the addition of Ni^{2+}, Hg^{2+}, Fe^{2+}, Co^{2+}, Ca^{2+}, Cd^{2+}, Mn^{2+}, Zn^{2+}, Cr^{3+}, Fe^{3+}, Al^{3+}, K^+, Na^+, Ag^+, respectively (Fig. 2a), which was consistent with the results of UV-Vis spectra (Fig. S4, ESI†). This observation indicated that **3** had high sensitivity and excellent selectivity for Cu^{2+} in CH_3CN–HEPES buffer solution (1 : 1, v/v). The corresponding titration of **3** with Cu^{2+} is provided in Fig. 2b. Upon gradual addition of Cu^{2+}, the fluorescence intensity of the solution at 575 nm increased gradually. The fluorescence intensity enhanced almost 8-fold upon the addition of 1.0 equiv. of Cu^{2+}. When more Cu^{2+} was titrated, the fluorescence intensity showed negligible changes. For convenience, the

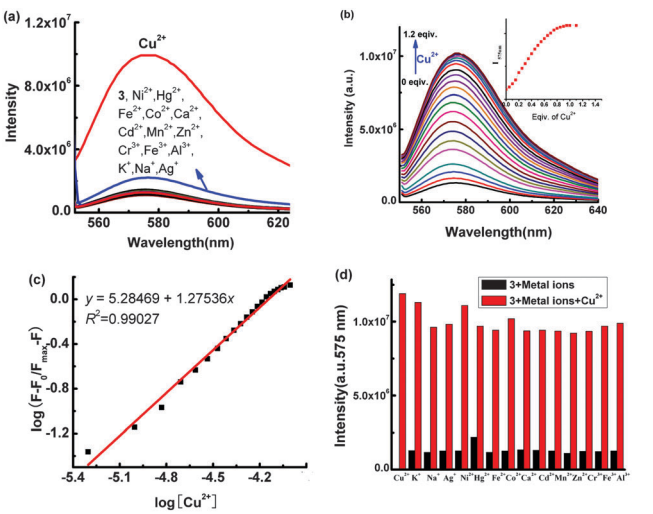


Fig. 2 (a) Fluorescence spectra of **3** (0.12 mg mL^{-1} **3**, equals 10 μM **2**) upon addition of 20 μM Cu^{2+} and other metal ions with the excitation at 540 nm in 1 : 1 CH_3CN–HEPES buffer solution (10.0 mM HEPES, pH 6.8). (b) Emission profile of **3** (10 μM) upon concomitant additions of Cu^{2+}. Inset: Changes of emission intensity at 575 nm with incremental addition of Cu^{2+}. (c) Hill plot of the fluorescence titration data of **3** at 575 nm with Cu^{2+} in CH_3CN–HEPES solution. (d) Fluorescence intensity changes of **3** (10 μM) upon addition of various metal ions (2.0 equiv.) in CH_3CN/HEPES (pH 6.8, v/v, 1 : 1, λ_{ex} = 540 nm, slit: 2.5/2.5 nm).

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concentration of **3** herein was changed to **2** coated on the surface (0.12 mg mL^{-1} **3** is equal to 10 μM **2**). From the Job's plot (Fig. S5, ESI†), a maximum fluorescence change was observed when the molar fraction of the sensor [**2**] *versus* [Cu^{2+}] + [**2**] was 0.5, indicating a 1 : 1 binding stoichiometry between **2** and Cu^{2+}, where [**2**] refers to the concentration of **2** coated on the surface of **3**. Based on a 1 : 1 binding mode, the binding constant for **2** and Cu^{2+} was calculated to be $\log K$ = 5.28 from the Hill plot of fluorescence titration (Fig. 2c) with satisfactory correlation coefficient values R = 0.990. The calculated detection limit for Cu^{2+} is (18.5 ± 0.6) nM using the equation $D_L = K \times Sb1/S$, where K = 3, Sb1 is the standard deviation of the blank solution (10 times) and S is the slope of the calibration curve.[35] The detection limit is much lower than the typical concentration of blood copper (13.7–23.6 μM) in normal individuals[36] and the limit of copper in drinking water (~20 μM) set by the U.S. Environmental Protection Agency.[37]

Competition experiments were also carried out by monitoring the change in the fluorescence intensity at 575 nm upon the addition of 2.0 equiv. of Cu^{2+} ions to a solution of **3** and different metal ions (10 equiv.) in the buffered solution (Fig. 2d). It indicates that none of the metal ions exert virtually any effect on the fluorescence intensity. In addition, we studied the emission behaviour of **3** treated with Cu^{2+} in different pH environments. In the pH range 5–8, a marked fluorescence enhancement was observed upon the addition of Cu^{2+}. The most strong fluorescence intensity appeared at around pH 6.8 for Cu^{2+}, which is close to physiological conditions (Fig. S6, ESI†). Therefore, the designed probe is capable of fluorescence recognition of Cu^{2+} with high selectivity and sensitivity.

It should be noted that the entire detection signals are from the rhodamine group. The NDs used herein are non-fluorescent and mainly act as matrices. The control experiment indicated that the fluorescence intensity of free **2** increased in the presence of Cu^{2+} without the nanodiamond core, as shown in Fig. S7 (ESI†). **2** and **3** have similar emission wavelengths. However, the enhanced fluorescence intensity of **2** is far less than that of **3** under the same conditions. It means that the conjugated ND probe **3** has high sensitivity for Cu^{2+} compared to the free **2**. This phenomenon is consistent with the references that the fluorescence intensity of a small probe will be enhanced after being connected to nanomaterials by preventing the inter-collision quenching of the fluorescent probe. Rhodamines display the brightest emission. However, a wide variety of molecules can act as collisional quenchers, such as water, oxygen, halogens, amines and fluorophores themselves. Besides collisional quenching, fluorescence quenching can occur by a variety of other processes. For example, fluorophores can form nonfluorescent complexes with quenchers. When the fluorophores are connected to nanomaterials, the chance of free collisional and the formation of nonfluorescent complexes in solution will be decreased greatly. As a result, the fluorescent intensity of **3** will be enhanced compared to the free **2**. That is the reason why we focus on the performance of **3** in this work.

3.2 NO reversible response by Cu(II)-based probe 3

Next, we performed UV-Vis and fluorescence titration studies of the **3** + Cu^{2+} (10 μM) system for NO at a time point of 30 min after the addition of DEA·NONOate. DEA·NONOate was used as the source of NO, which is commercially available with a half time of 16 min. When 80 μM DEA·NONOate was added into the **3**–Cu^{2+} system, the fluorescence at 575 nm was quenched dramatically (Fig. 3a). It indicated that the fluorescence of the **3**–Cu^{2+} system was quenched when the Cu(II) centre was reduced to Cu(I) species *via* NO-triggered reduction.[7] The coordination of the **3**–Cu^{2+} system was broken and the spirolactam ring returned to the closing state. Meanwhile, **3** showed weak absorption at 555 nm (Fig. 3b). Upon the addition of Cu^{2+}, selective absorption 'turn-on' changes were observed at 555 nm with a 10-fold enhancement. However, when 80 μM NO was added into the **3**–Cu^{2+} system, the absorption intensity decreased severely. Except NO, other agents such as F^-, Cl^-, Br^-, I^-, CO_3^{2-}, SO_3^{2-}, NO_2^-, S^{2-} and ClO^- induced a relatively small number of changes. Competition experiments were also carried out by monitoring the change in absorption spectra at 555 nm upon the addition of 2.0 equiv. of NO to a solution of the **3**–Cu^{2+} system and different metal ions (10 equiv.) in the buffered solution (Fig. S8, ESI†). It can be observed that none of these agents exerts virtually any effect on the agents.

The corresponding titration of **3**–Cu^{2+} with NO is provided in Fig. 3c. Upon the gradual addition of NO, the fluorescence intensity of the solution at 575 nm decreased gradually. An excellent linear correlation (R^2 = 0.994) between the fluorescence intensities and the DEA·NONOate concentrations from 0 to 60 μM was observed (Fig. 3c), and the detection limit was estimated to be (10.5 ± 0.4) nM based on $D_L = K \times Sb1/S$. It is lower than previous studies.[12] Therefore, the probe **3**–Cu^{2+}

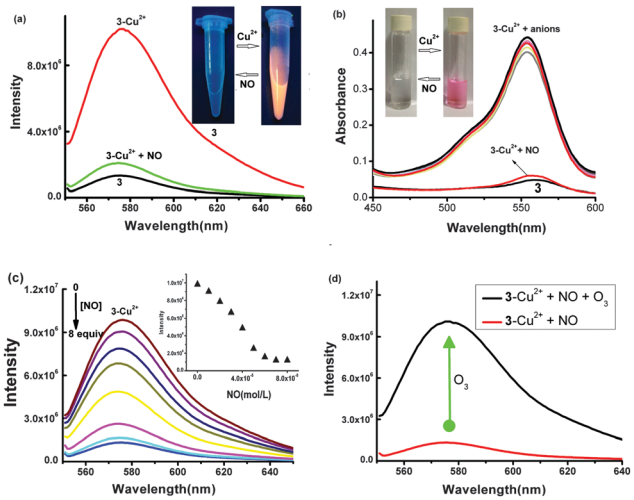


Fig. 3 (a) Fluorescence spectra of **3**–Cu^{2+} ([Cu^{2+}] = 10 μM) treated with DEA·NONOate (80 μM) in CH_3CN/HEPES (pH 6.8). Inset: the changes of fluorescence emission of **3** in the presence of Cu^{2+} and NO. (b) Absorption spectra of **3** (0.12 mg mL^{-1}) + Cu^{2+} (10 μM) in the presence of different anions and DEA·NONOate (80 μM F^-, Cl^-, Br^-, I^-, CO_3^{2-}, SO_3^{2-}, NO_2^-, S^{2-}, ClO^- or NO) in CH_3CN/HEPES buffered solution at pH 6.8. Inset: the color changes of **3** in the presence of Cu^{2+} and NO. (c) Emission profile of **3**–Cu^{2+} ([Cu^{2+}] = 10 μM) upon concomitant additions of NO in CH_3CN/HEPES. Inset: Changes of emission intensity at 575 nm with incremental addition of NO. (d) Fluorescence spectra of **3**–Cu^{2+} (10 μM) + NO (80 μM) in the addition of O_3 in 10 mM CH_3CN/HEPES (1 : 1, v/v; pH 6.8, λ_{ex} = 540 nm).

is capable of fluorescence recognition of NO with high selectivity and sensitivity.

If the Cu(II) centre is reduced to Cu(I), it is thought that this reaction can be reversed when some oxidant is added. To confirm this, superoxide O_3 was employed as the oxidant, which is shown in Fig. 3d. As expected, when an appropriate volume of O_3 solution was added into the **3**–Cu^{2+} + NO system in CH_3CN/HEPES, the fluorescence intensity at 575 nm was restored quickly within 1.0 min. It indicated that Cu(I) had been oxidized to Cu(II) by superoxide and the fluorescence of **3** was triggered again. So the recognition of NO by **3**–Cu^{2+} is a reversible process (Fig. 1). On the other hand, the 'On–Off' changes of the **3**–Cu^{2+} + NO system may be developed as the first reversible fluorescence probe for O_3. Ozone exposure is a growing global health problem, especially in urban areas. Moreover, it has recently been shown that ozone may be produced endogenously due to inflammation and antibacterial responses of the immune system.[38] Its detection in the environment and cells is also important because of its toxic effects.[39] However, ozone's role in damaging human tissues is unclear, so there is a need for a biomedical probe to better understand the localization of this highly reactive species. It is worth noting that the detection of ozone in biological and atmospheric samples is rarely reported, because interference from different reactive oxygen species is often a problem.[40,41] It is well known that Cu(I) is relatively stable in CH_3CN in the atmosphere; the oxidizing ability of O_2 is far more weak than O_3. P. Faller reported that the oxidation of Cu(I) by O_2 is a slow process.[42] Therefore, the **3**–Cu^{2+}–NO system may be developed as a fluorescent probe for the atmospheric detection of O_3. These results show that further studies on the intracellular biological activity of O_3 are warranted.

3.3 Stability and reusability of probe 3

The synthetic routes toward **2** and **3** are illustrated in Scheme 1. Nanoparticles **3** were produced *via* the formation of an ester bond between the hydroxyl group of rhodamine **2** and the ND surface functionalized with thionyl chloride of **1**, which is depicted in Scheme 1. The amount of **2** adsorbed onto the surface of NDs was determined by calculating the change in **2** concentrations before and after adsorption using a UV-Vis spectrophotometer at 555 nm according to the obtained standard equation (Fig. 4a). The amount of **2** conjugated was determined to be nearly (50 ± 6) μg per milligram NDs, that is, 0.12 mg mL^{-1} **3** was coated with 10 μM **2**.

Then the storage stability study of **2** and **3** was performed in CH_3CN/HEPES (pH 6.8) solution at room temperature for 100 h, as shown in Fig. 4b. The fluorescence intensity of the free rhodamine **2** at 575 nm gradually increased with time. Compared with **2**, **3** exhibited excellent stability under the same conditions. It is normally deemed that the spirolactam ring is unstable and resulted in a slight opening-state by the inter-collision or thermal motion in solution,[43] while the conjugation to NDs decreases the inter-collision quenching of **2**.

It should be pointed out that the interaction between **2** and ND include covalent conjugation and physical adsorption.

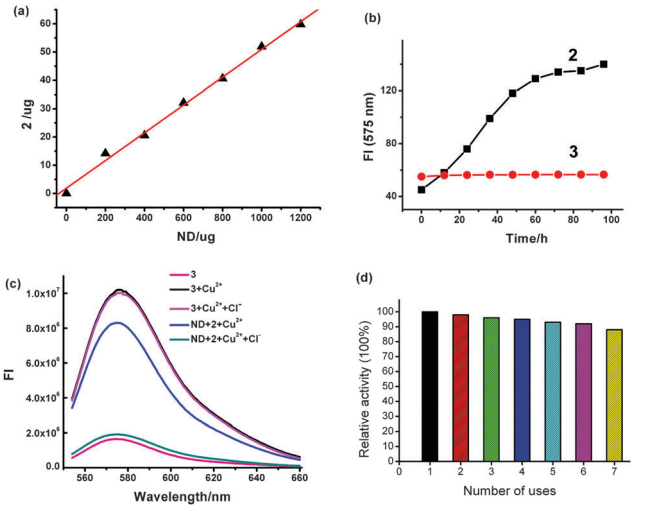


Fig. 4 (a) The standard curve of ND bound **2**. (b) Storage stability of **2** (10 μM) and **3** (10 μM) in CH_3CN/HEPES with time. (c) The fluorescence spectra of **3** (10 μM), **3** + Cu^{2+} (10 μM) and the ND + **2** + Cu^{2+} (10 μM) system before and after the addition of a chloride salt in CH_3CN/HEPES solution. When the chloride salt was added, the suspension was centrifuged. The precipitate was redissolved in fresh CH_3CN/HEPES solution. (d) Reusability of **3**. Reusability of **3** sensing Cu^{2+} and NO was monitored for six successive days. **3** was taken in triplicates and was assayed for the remaining percent activity. Activity determined on the first day was taken as control (100%) for the calculation of remaining activity after each use (λ_{em} = 575, λ_{ex} = 540 nm, pH 6.8).

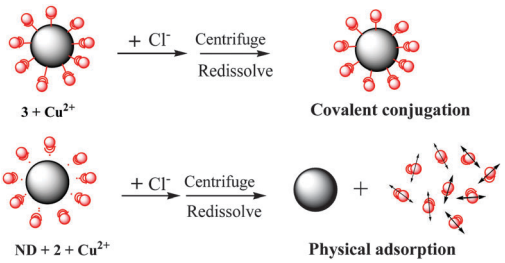
In order to confirm the covalent bond between **2** and ND, the effect of excess chloride salt on the fluorescence intensity of **3** + Cu^{2+} and ND + **2** + Cu^{2+} system was investigated, as shown in Fig. 4c. To be specific, when excess chloride salt was added, the suspension was centrifuged and the supernatant was removed. The precipitate was washed for three times in order to remove salts and redissolved in CH_3CN/HEPES solution again. The fluorescence was monitored before and after the addition of excess chloride salt was measured. As can be seen from Fig. 4c, the fluorescence of precipitate **3** + Cu^{2+} (black line) would not be affected by the addition of excess chloride salt (pink line). However, when excess chloride salt was added into the ND + **2** + Cu^{2+} system (blue line), the fluorescence of its precipitate was quenched severely (green line). It indicates that the interaction between **2** and ND is weakly physical adsorption in the ND + **2** + Cu^{2+} system, and **2** could be removed from the surface of ND by chloride salt (salt effect[44,45]), while the interaction between **2** and ND onto **3** is the covalent bond, it is tight enough for the medium interaction. For clarity, interaction of chloride salt was shown in Scheme 2.

Furthermore, reusability of **3** sensing Cu^{2+} and NO was monitored for six successive days (Fig. 4d). Sensing ability of **3** showed 92% and 88% of the initial activity, after its 5th and 6th repeated use, respectively. The good reusability is not only due to the easy separation of the **ND** probe from the reaction mixture, but also due to the tight covalent bond interaction just motioned above.

3.4 Biocompatibility and cytotoxicity of 3

In order to determine whether the ND probe can be used in studies related to the NO's role in cell biology we incubated

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Scheme 2 Schematic drawing of interaction of the chloride salt and **3** + Cu^{2+}, ND + **2** + Cu^{2+}.

SW480 cells with compound **3**. Confocal fluorescence images of Cu^{2+} and NO in SW480 cell lines were measured. Briefly, SW480 cells were incubated with **3** (10 μM) in DMEM medium for 60 min at 37 °C and washed with PBS buffer (pH 6.8) to remove excess **3**. The results are shown in Fig. 5.

As determined by fluorescence imaging microscopy, stained cells treated with 10 μM solution of **3** for 60 min remained weakly fluorescent within the cells (Fig. 5b). After treatment with Cu^{2+} ions (20 μM) for further 30 min, the cells pretreated with **3** (10 μM) obviously showed a significant fluorescence increase from the intracellular region (Fig. 5c). However, once 160 μM NO was added and incubated for another 30 min, the red fluorescence disappeared (Fig. 5d). These results suggest that **3** is cell permeable with low cytotoxic effect and an effective intracellular Cu^{2+} ion and a NO imaging agent with "OFF–ON" red colour fluorescence emissions. This new Cu^{2+} and NO selective fluorescent probe may have potential bio-medical applications.

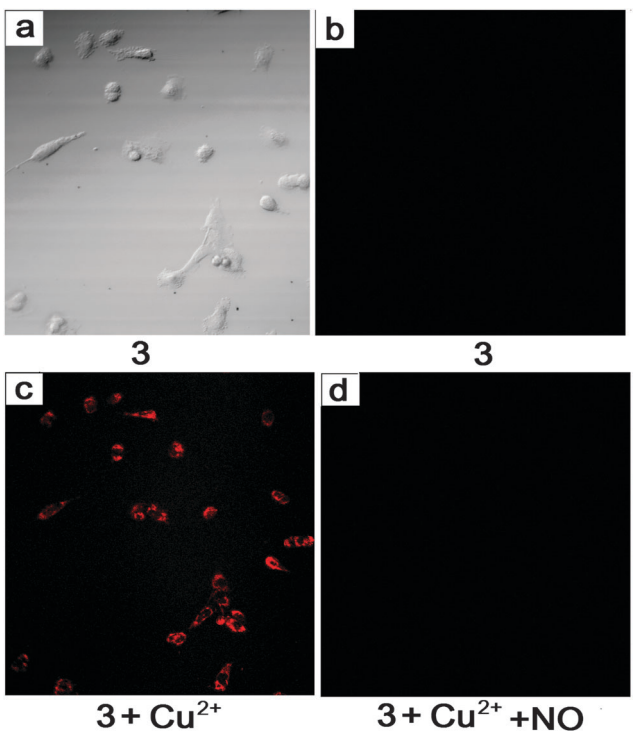


Fig. 5 Images of SW480 cells: (a) bright field image of cells incubated with **3** (10 μM) for 60 min; (b) fluorescence image of (a); (c) fluorescence image of (b) in the presence 20 μM Cu^{2+} for 30 min at 37 °C. (d) Fluorescence image of (c) in the presence 160 μM NO for another 30 min. Emission was collected at 580–630 nm for the red channel (excited at 543 nm) by using a [10×] objective.

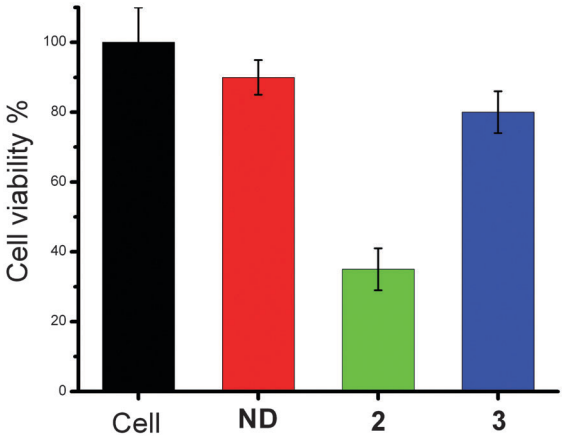


Fig. 6 The viability assay of cell alone, with ND (0.12 mg mL^{-1}), **2** (10 μM) or **3** (10 μM) on SW480 cells using the MTT assay for 12 h. All samples were done in triplicate.

It is well known that the rhodamine B dye has obvious cytotoxicity to cells. To further evaluate the practical applicability of the designed chemosensor ND probe **3**, cytotoxicity was determined on SW480 cells using the MTT assay for 12 h; the results are shown in Fig. 6. Generally, no significant differences in the proliferation of the cells were observed in NDs alone, which showed the excellent biocompatibility of ND nanoparticles (90%). Notably, free **2** exhibited the severest cytotoxicity (35%). It is comforting to see that the cytotoxicity of **2** is decreased remarkably when conjugated to NDs, the cell viability reached 82% for **3**. The existence of NDs is beneficial for reducing the cytotoxicity of **2**. As such, this study provides a convenient method to improve the detection efficiency and decrease the cytotoxicity of rhodamine B probes, which further extends the applicability of the rhodamine dye in living cells.[46]

4. Conclusion

In this study, we have developed a fluorescent probe, where Cu(II)-based rhodamine B was covalently bound to the nanodiamond (ND) for the determination of NO. Herein NDs act as highly efficient matrices and rhodamine B acts as a fluorescent group. NDs provide a greater surface area for binding a higher amount of probes and prevent inter-collision quenching of the rhodamine B probe. It offers greater selectivity and sensitivity to Cu^{2+} and NO sequentially than free rhodamine B. Such a ND probe is also useful for the study of O_3. Compared with the free rhodamine B, other advantages of the ND probe involve its reusability (as observed by reusability experiment), enhanced stability against chemical denaturants (sodium chloride), good biocompatibility (cell images) and lower cytotoxicity (MTT assay). This new Cu^{2+} and NO selective fluorescent ND probe may have potential applications in environmental science and biology.

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